

Checkout Intelligence Agent

Product Requirements Document

Team Seven Up · Version 1.0 · 2026

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1. Executive Summary

The Checkout Intelligence Agent is an AI-powered commerce platform designed to reduce cart abandonment and recover lost revenue in real time. When a shopper exhibits hesitation signals during checkout — idling on a payment page, repeatedly changing shipping options, failing to apply a coupon, or moving toward exit — the system automatically intervenes with a conversational AI assistant (KasparAI) that diagnoses the friction and offers targeted incentives to close the sale.

The platform combines behavioral analytics, a multi-agent AI backend, and a polished React storefront to deliver measurable lift in conversion rates without disrupting the customer experience.

Key Metric	Target
Cart Abandonment Reduction	≥ 15%
AI Intervention Accuracy	> 55% confidence threshold
Recovery Conversation Latency	< 2 seconds (p95)
Friction Score Resolution	100-point scale, trigger at ≥ 50

2. Problem Statement

E-commerce cart abandonment rates industry-wide sit between 65–80%. The primary reasons shoppers abandon checkout include unexpected costs, payment friction, distrust, and decision paralysis. Traditional solutions — retargeting emails, discount pop-ups — are reactive, impersonal, and arrive too late.

- Price Hesitation:** Shopper stalls on pricing or hovers over totals, uncertain about value.
- Shipping Confusion:** Repeated toggling between shipping methods indicates uncertainty about delivery timelines or costs.
- Coupon Confusion:** Failed discount code attempts signal expectation of a deal that hasn't been communicated.
- Exit Intent:** Cursor movement toward browser chrome or sudden inactivity signals imminent abandonment.
- Payment Errors:** Card failures or unfamiliar payment forms cause drop-offs at the last step.
- Trust Gaps:** Absence of security signals or return policy clarity reduces buyer confidence.

3. Product Goals & Success Criteria

Goal	Description	Success Metric
Real-time Friction Detection	Identify checkout hesitation as it happens	Friction score $\geq$ 50 within 3 events
Contextual AI Intervention	Trigger KasparAI with the right message at the right moment	$\geq$ 55% classifier confidence
Conversational Recovery	Resolve friction through natural dialogue and targeted offers	RESOLVED token emitted
Discount & Offer Management	Dynamic discount codes based on cart value and friction type	Complete offer tier applied
Analytics Visibility	Dashboard showing recovered revenue, interventions, and abandoned orders	Dashboard populated in real time
Shopify Integration	Sync real product catalog via GraphQL Admin API	Products load in < 1s with fallback

4. User Stories

ID	Role	Story	Acceptance Criteria
US-01	Shopper	I want to receive help when I'm unsure about pricing so I don't abandon my cart	Kaspar AI appears with a helpful offer when friction score $\geq$ 50 and friction type is detected
US-02	Shopper	I want clear shipping options so I don't abandon my cart	Agent delivers timely, relevant offers, and optionally unlocks free express shipping with a qualifying offer
US-03	Shopper	I want to use a valid discount code so I feel rewarded	Agent suggests SAVE10 or WELCOME15 when a coupon attempt fails; valid codes are accepted
US-04	Shopper	I want to feel secure before entering my card details	Agent proactively surfaces SSL encryption and 30-day return guarantee
US-05	Store Admin	I want to see how many carts were saved by the agent	Analytics dashboard shows % of total interventions, recovered orders, and revenue
US-06	Store Admin	I want the agent to use my live Shopify catalog so I can see real products	Products are fetched via Shopify GraphQL API with graceful mock fallback

5. Feature Specification

5.1 Friction Scoring Engine

- Every behavioral event (idle, coupon failure, exit intent, shipping toggle, etc.) increments a friction score on a 0–100 scale.
- Weighted event map: exit_intent +30, coupon_failed +25, idle_checkout +20, shipping_method_changed +15, price_hover +15, cart_modified +10, address_edited +10.
- Score is capped at 100 and persisted in MongoDB per session.
- When score ≥ 50 , the AI hesitation analyzer is invoked.

5.2 AI Hesitation Classifier

- Passes full event log and session context to GPT-4o-mini with structured JSON response format.
- Returns one of: price hesitation, shipping confusion, coupon confusion, exit intent, payment error, trust gap, other.
- Confidence threshold of 0.55 required before triggering intervention.
- Rule-based fallback activates when OpenAI API key is absent or call fails.

5.3 KasparAI Chat Widget

- Floating chat UI rendered in the bottom-right corner of the checkout page.
- Opens automatically when an intervention is triggered with a friction-type-specific opening message.
- Quick-reply shortcuts for common needs: 10% Off, Free Shipping, Payment Help, Return Policy.
- Proactive message rotation every 10 seconds when widget is minimised (badge + pulse animation).
- Snackbar pre-nudge shown before full widget appears on first proactive tick.
- Unread message counter displayed on minimised widget header.
- Exit-intent listener triggers the agent when cursor leaves viewport top.

5.4 Multi-Agent Recovery System

- CheckoutAgent: Analyzes navigation patterns and idle behavior; provides contextual persuasion copy.
- PricingAgent: Generates tiered discount offers — SAVE10 (10% for orders > \$0) and WELCOME15 (15% for orders > \$500); validates coupon codes.
- ShoppingAgent: Recommends products from the same category as cart items; powers keyword-based product search.
- The active agent is selected dynamically based on keywords in the user's message and current checkout stage.

5.5 Offer & Coupon Management

- Valid codes: SAVE10 (10% off), WELCOME15 (15% off), FREESHIP (free standard shipping).
- Coupon validation is performed client-side for immediate feedback and server-side for persistence.
- Applied offers are reflected in the order summary with itemised discount lines.
- Remove-item modal intercept: when a shopper tries to delete a cart item, an offer modal appears first.

5.6 Analytics Dashboard

- Displays: Saved Revenue, Total Interventions, Conversion Lift %, Abandonment Rate.
- Bar chart showing revenue recovered over time (last 30 days / this week filter).
- Breakdown of top intervention types with action taken and percentage share.
- Powered by aggregation queries on the MongoDB sessions collection.

6. User Experience Flows

Primary Checkout & Recovery Flow

Shopper adds items to cart → navigates to /checkout → behavioral events are logged in real time → friction score accumulates → at threshold ≥ 50 , AI classifier runs → KasparAI widget opens with targeted message → shopper engages in conversation → agent applies offer → [RESOLVED] token sent → session marked complete → order confirmed.

Exit Intent Interception

Cursor moves above viewport → exit_intent event fires → friction score jumps → KasparAI opens immediately with exit-specific message and 10% discount offer.

Coupon Failure Recovery

Shopper enters invalid code → coupon_failed event fires → agent widget opens → agent suggests SAVE10 or WELCOME15 → shopper applies valid code → checkout proceeds.

Remove Item Retention

Shopper clicks remove item → intercept modal appears → agent offers 10% discount → shopper keeps item with discount applied → modal dismissed.

7. Non-Functional Requirements

Category	Requirement
Performance	Chat API response within 2s (p95); friction score update within 500ms
Availability	Backend target 99.5% uptime; graceful degradation with rule-based fallback
Security	256-bit SSL on all checkout pages; no payment card data stored server-side
Scalability	MongoDB session store supports horizontal scaling; stateless Express API
Privacy	Session IDs are UUIDs; no PII stored in event logs; compliant with GDPR principles
Browser Support	Chrome 100+, Firefox 100+, Safari 15+, Edge 100+; responsive mobile layout

8. Out of Scope

- Actual payment processing (the checkout form is a UI prototype; no real Stripe charges are made).

- Email retargeting or SMS follow-up after session expiry.
- Multi-language support (English only in v1).
- Native mobile applications (web-responsive only).
- Admin panel for configuring friction thresholds or offer codes without code changes.

9. Product Roadmap

Phase	Milestone	Status
Phase 1	Core friction scoring + rule-based fallback agent	✓ Complete
Phase 2	GPT-4o-mini classifier + conversational recovery	✓ Complete
Phase 3	Multi-agent architecture (Shopping, Pricing, Checkout agents)	✓ Complete
Phase 4	Analytics dashboard + Shopify GraphQL integration	✓ Complete
Phase 5	EMI / BNPL payment options, A/B testing of offer strategies	Planned
Phase 6	Admin configuration UI, webhook-based order sync	Planned